

Cover letter — *Nature Machine Intelligence*

Dear Editors,

We submit for your consideration “**A sustained one-dimensional channel underlies in-context learning in autoregressive Transformers**” (9 main figures + 1 Extended Data, 4,000-word body, full source data deposited).

The finding, distinct from concurrent head-level accounts. Few-shot in-context learning (ICL) in causal Transformers operates through a single *layer-spanning* one-dimensional channel of the residual stream — the *sustained causal bottleneck* (SCB). Concurrent work characterises a *Hidden Geometry of ICL* (Yang & Zhao 2025) at the head/component level, and individual function-vector heads have been studied (Cabannes et al. 2025); we operate one level below, at the residual-stream **layer** level into which those head-level mechanisms write. We demonstrate three properties absent from the head-level descriptions: (i) eigenvector continuity ≥ 0.99 across the SCB band of 8–29 consecutive layers, (ii) residual-stream projection-out causal control with a clean if-and-only-if cluster (five SCB-free causal LMs at baseline $R \approx 1$: BLOOM-1B1, Qwen2.5-0.5B, Qwen2.5-1.5B, GPT-Neo-2.7B, OLMo-7B), and (iii) reproducibility under independent re-experiment ($n_col\ 11 \rightarrow 0$ across 6/6 seeds; raw λ_1/λ_2 collapses from 682.6 to mean 2.40, \$ 285\times\$ drop). Across foundation models in four modalities, the SCB length scales linearly with ICL amplification (Pearson $r = 0.836$, $p = 2.9 \times 10^{-5}$, $n = 17$ SCB-bearing deep-bootstrap panel).

Causal control is asymmetric and reproducible. Spectral-regularised full-parameter fine-tuning destroys the channel on Pythia-1B ($n_col\ 11 \rightarrow 0$ across 6/6 seeds; raw λ_1/λ_2 from 682.6 to mean 2.40) but cannot retrofit one onto the SCB-free BLOOM-1B1 ($n_col\ 15 \rightarrow 15$ across 10/10 seeds and both spectral-coefficient settings $\alpha \in \{0.1, 1.0\}$; the regulariser does inflate the within-band λ_1/λ_2 amplitude by $+66 \pm 3$ at $\alpha=0.1$ and $+227 \pm 5$ at $\alpha=1.0$ on a baseline of 283, but no new layer-spanning direction emerges). Five SCB-free causal LMs (BLOOM-1B1, Qwen2.5-0.5B, Qwen2.5-1.5B, GPT-Neo-2.7B, OLMo-7B) all sit at $R \approx 1$, an if-and-only-if control.

Cross-modality test of the V_eff theory. We directly probed 60 foundation models with full participation-ratio profiles (cycle-32 baseline + cycle-74 LEO5 + MUSICA-INN re-aggregated). The data confirm a four-class V_eff prediction: (i) all 17 SCB-bearing causal LMs (high V_eff + temporal asymmetry) form SCBs; (ii) 15/15 text/protein encoder MLMs (high V_eff , no temporal asymmetry) sit at $n_col = 0$; (iii) both Nucleotide Transformer scales (DNA 6-mer tokenizer, $V_eff \approx 4096$) form a strong SCB ($n_col = 18, 24$, 3-seed reproduced); (iv) four of six vision Transformers (low effective patch entropy) form SCBs, including ViT-base ($n_col = 8$ across $L = 12$) and DINOv2-large ($n_col = 6$). The encoder–decoder dichotomy is not the discriminator — what predicts the SCB is a low-rank token distribution combined with the causal training objective, or either condition alone in extreme cases. This sharpens the original “SCB-exclusive-to-AR” framing into an empirical V_eff law that spans text, protein, DNA and vision modalities.

From-scratch saturation test (Application B). To probe the V_eff prediction causally rather than only correlationally, we ran an equal-architecture, equal-compute from-scratch pretraining sweep on a 160-M Pythia-style decoder (FineWeb-edu, 5-B-token budget, AdamW, cosine schedule), varying only the tokenizer vocabulary across four orders of magnitude: $|V| \in \{256, 1024, 8192, 50257, 100000, 200000\}$ with 4–8 independent seeds per cell (35 5-B-token runs total; a fused-cross-entropy kernel removes the single-H100 logits-tensor OOM at $|V| \geq 10^5$, implementation-only, no methodological change). All four mature-ICL cells ($|V| \in \{8192, 50257, 100000, 200000\}$) collapse onto the saturation band: per-cell mean $n_col \in [6.57, 7.12]$, all 27 mature seeds clear the maturity

threshold $R \geq 1.5$ (range [2.243, 2.895]), and a one-way ANOVA across the four mature cells gives $F(3, 23) = 0.90$, $p = 0.46$ — saturation across more than a 24-fold range in $|V|$ at fixed architecture and compute. The two non-mature cells behave as the theory permits: $|V| = 1024$ sits on the rising edge of ICL maturity (mean $R = 1.65$, two of four seeds below threshold) and $|V| = 256$ has not reached mature ICL within 5 B tokens (mean $R = 1.10$), so the saturation prediction is not directly tested at those points (conditional on mature ICL). Application B therefore lifts the original three-vocabulary correlational test into a six-vocabulary causal test of the saturation magnitude at $L \cdot 0.6 \approx 7$.

Biomedical few-shot benchmark (Application D). A direct, non-trivial implication of the V_{eff} theory: biomedical-specialised pretraining narrows the effective vocabulary distribution and should therefore *not* dominate matched-scale general-purpose pretraining on biomedical few-shot QA. We tested this on PubMedQA and MedQA-4opt (default zero-shot lm-eval-harness protocol) with four causal LMs at matched scale. The general-purpose Pythia-2.8B *outperforms* both biomedical-specialised models on PubMedQA (0.620 vs. BioGPT-Large 0.480 and BioMedLM 0.418); the in-domain advantage on MedQA-4opt is at most marginal (BioMedLM 0.273 vs. Pythia-1B 0.263 vs. Pythia-2.8B 0.246, all near the four-option random baseline 0.25). The pattern is consistent with a V_{eff} -bottleneck reading and is reported as an empirical implication rather than a headline benchmark; a Llama-3.2-1B point is queued and will appear in the camera-ready revision.

Why NMI is the right venue. The work delivers two concrete machine-intelligence advances: (i) a precise *diagnostic* — SCB length predicts mature ICL behaviour better than parameter count alone (LOO $R^2 = 0.62$ vs -0.02), at two to three orders of magnitude lower compute than a full ICL benchmark; (ii) a *V_{eff} -bottleneck* theory linking the gradient row-space rank to the 1D-channel limit, with cross-modality predictions empirically confirmed across 60 pre-trained foundation models in text, protein, DNA and vision modalities, causally tested via the Application B six-vocabulary from-scratch sweep, and with a non-trivial biomedical few-shot implication confirmed in Application D. Total compute is \$150 H100-equivalent hours (Methods §M.compute) — \$25 hours for cross-architectural probing of pre-trained models, \$115 hours for the Application B 35-run from-scratch sweep across all six vocabularies, and \$10 hours for the Application D biomedical-eval matrix.

Reproducibility. All probe scripts, aggregation code, figure generators, and the full set of source-data files accompany the submission, including the 18 newly run cycle-74 PR-profile JSONs that anchor the encoder-null claim with direct measurements (rather than aggregated archive numbers), the 35 per-cell summary JSONs for Application B, and the 8 model $\times$ task accuracy JSONs for Application D. Code repository: <https://github.com/Jie-Ni/biolm-scb>. Limitations — including probe-protocol divergence between all-position and last-token aggregations, and the heuristic (not closed-form) status of §S6 theory — are explicitly disclosed in Methods §M.4 and a dedicated Limitations paragraph.

Suggested reviewers (5; see `suggested_reviewers_v4_SCB.md`): Trenton Bricken (Anthropic), Christopher Olah (Anthropic), Jacob Andreas (MIT CSAIL), Klaus-Robert Müller (TU Berlin), Atticus Geiger (Pr(Ai)²R Group).

We declare no conflicts of interest; the work was conducted on the LEO5 cluster (University of Innsbruck) and the Austrian Scientific Computing MUSICA clusters with no external funding.

Sincerely,

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